Xiang Gu

Assistant Professor, Department of Information Science
School of Mathematics and Statistics, Xi'an Jiaotong University
Xi'an, Shaanxi 710049, P. R. China

xianggu@xjtu.edu.cn
xjtu-xgu.github.io/xianggu

Academic Appointments

- Assistant Professor, Department of Information Science, School of Mathematics and Statistics, Xi'an Jiaotong University.

Education

- Ph.D. in Applied Mathematics, Xi'an Jiaotong University, 2024.
  Advisors: Prof. Zongben Xu; Co-advisor: Prof. Jian Sun.
- B.S. in Statistics, Southeast University, 2017.

Research Interests

Generative models; Optimal transport theory and algorithms; AI for science.

Publications

*(# indicates corresponding author; * indicates co-first author.)*

Journal Articles

J1. **Xiang Gu**, Yucheng Yang, Wei Zeng, Jian Sun, Zongben Xu. "Keypoint-Guided Optimal Transport: Models, Algorithms, and Applications." *Journal of Machine Learning Research (JMLR)*, 2026.

J2. **Xiang Gu**, Huichun Zhang, Jian Sun. "A Residual-Shell-Based Lower Bound for Ollivier–Ricci Curvature." *Applied Mathematics Letters*, 2026.

J3. Liwei Yang, **Xiang Gu**[#], Shipeng Wang, Jian Sun. "Learning Domain Generalizable Semantic Segmentation Model by Transformation-Invariant Feature Subspace Projection." *Neurocomputing*, 2026.

J4. Yufei Luo, **Xiang Gu**, Jian Sun. "Generative Flow Model on Distance Geometry for Predicting Transition States of Chemical Reactions." *Nature Communications*, 2026.

J5. Xiaole Tang, Xiaoyi He, Jiayi Xu, **Xiang Gu**, Jian Sun. "Learning Continuous Wasserstein Barycenter Space for Generalized All-in-One Image Restoration." *IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026.

J6. Yannan Pu, **Xiang Gu**, Niansheng Tang, Jingyao Zhang, Jian Sun. "Variational Masking Generative Model for Anomaly Detection on Incomplete Tabular Data." *Pattern Recognition*, 2026.

J7. Liwei Yang, **Xiang Gu**[#], Jian Sun. "In-Context Learning Implicit Representation for Time Series Forecasting." *IEEE Signal Processing Letters*, 2025.

J8. Jiazhen Wang, Zhihao Shi, **Xiang Gu**, Yan Yang, Jian Sun. "Diffusion-based Arbitrary-Scale Magnetic Resonance Image Super-Resolution via Progressive K-space Reconstruction and Denoising." *Medical Image Analysis (MIA)*, 2025.

J9. Xiaole Tang, **Xiang Gu**, Xiaoyi He, Xin Hu, Jian Sun. "Degradation-Aware Residual-Conditioned Optimal Transport for Unified Image Restoration." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025. (CCF-A)

J10. Xin Wei, **Xiang Gu**, Jian Sun. "Multi-scale Part-based Feature Representation for 3D Domain Generalization and Adaptation." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2024. (CCF-A)

J11. **Xiang Gu**, Xi Yu, Yan Yang, Jian Sun, Zongben Xu. "Adversarial Reweighting with α-Power Maximization for Domain Adaptation." *International Journal of Computer Vision (IJCV)*, 2024. (CCF-A)

J12. Xi Yu, **Xiang Gu** *, Jian Sun. "Contrasting Augmented Features for Domain Adaptation with Limited Target Domain Data." *Pattern Recognition*, 2023.

J13. **Xiang Gu**, Jian Sun, Zongben Xu. "Adversarial Data Splitting for Domain Generalization." *Science China Information Sciences (SCIS)*, 2023. (CCF-A)

J14. Fan Wang, **Xiang Gu**, Jian Sun, Zongben Xu. "Learning-based Local Weighted Least Squares for Algebraic Multigrid Method." *Journal of Computational Physics (JCP)*, 2023.

J15. **Xiang Gu**, Jian Sun, Zongben Xu. "Unsupervised and Semi-supervised Robust Spherical Space Domain Adaptation." *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2022. (CCF-A)

Conference Papers

C1. Taoran Zheng, Xing Li, Yan Yang, **Xiang Gu** #, Zongben Xu, Jian Sun. "Towards Prospective Medical Image Reconstruction via Knowledge-Informed Dynamic Optimal Transport." *NeurIPS*, 2025. (CCF-A)

C2. Dongyi Wang, Yuanwei Jiang, Zhenyi Zhang, **Xiang Gu** #, Peijie Zhou, Jian Sun. "Joint Velocity-Growth Flow Matching for Single-Cell Dynamics Modeling." *NeurIPS*, 2025. (CCF-A)

C3. Xi Yu, **Xiang Gu** #, Zhihao Shi, Jian Sun. "Wasserstein Style Distribution Analysis and Transform for Stylized Image Generation." *ICCV*, 2025. (CCF-A)

C4. Xiaole Tang, Xin Hu, **Xiang Gu**, Jian Sun. "Residual-Conditioned Optimal Transport: Towards Structure-Preserving Unpaired and Paired Image Restoration." *ICML*, 2024. (CCF-A)

C5. **Xiang Gu**, Liwei Yang, Jian Sun, Zongben Xu. "Optimal Transport-Guided Conditional Score-Based Diffusion Model." *NeurIPS*, 2023. (CCF-A)

C6. Xi Yu, **Xiang Gu** *, Haozhi Liu, Jian Sun. "Constructing Non-isotropic Gaussian Diffusion Model Using Isotropic Gaussian Diffusion Model." *NeurIPS*, 2023. (CCF-A)

C7. Zixiang Zhao, Jiangshe Zhang, **Xiang Gu**, Chengli Tan, Shuang Xu, Yulun Zhang, Radu Timofte, Luc Van Gool. "Spherical Space Feature Decomposition for Guided Depth Map Super-Resolution." *ICCV*, 2023. (CCF-A)

C8. Liwei Yang, **Xiang Gu** *, Jian Sun. "Generalized Semantic Segmentation by Self-supervised Source Domain Projection and Multi-level Contrastive Learning." *AAAI*, 2023. (CCF-A)

C9. Yucheng Yang, **Xiang Gu** *, Jian Sun. "Prototypical Partial Optimal Transport for Universal Domain Adaptation." *AAAI*, 2023. (CCF-A)

C10. Xin Wei, **Xiang Gu**, Jian Sun. "Learning Generalizable Part-based Feature Representation for 3D Point Clouds." *NeurIPS*, 2022. (CCF-A)

C11. **Xiang Gu**, Yucheng Yang, Wei Zeng, Jian Sun, Zongben Xu. "Keypoint-Guided Optimal Transport with Application in Heterogeneous Domain Adaptation." *NeurIPS*, 2022. (CCF-A)

C12. **Xiang Gu**, Xi Yu, Yan Yang, Jian Sun, Zongben Xu. "Adversarial Reweighting for Partial Domain Adaptation." *NeurIPS*, 2021. (CCF-A)

C13. **Xiang Gu**, Jian Sun, Zongben Xu. "Spherical Space Domain Adaptation with Robust Pseudo-label Loss." *CVPR*, 2020.

Honors & Awards

- CAAI Outstanding Doctoral Dissertation Award.
- Shaanxi "Sanqin Bochuang" Talent Program.
- National Postdoctoral Innovation Talent Support Program.
- NSFC Young Student Basic Research Program (PhD Student Project).
- VALSE Outstanding Student Forum Guest Representative.

Academic Service

- **Editorial Board**: Journal of Mathematical Learning and Computation.
- **Area Chair**: ICLR (2027).
- **Reviewer (Journals):** IEEE Transactions on Pattern Analysis and Machine Intelligence; Journal of the American Statistical Association; International Journal of Computer Vision; IEEE Transactions on Image Processing; etc.
- **Program Committee:** CVPR, ICCV, NeurIPS, ICLR, ICML, ECCV, AAAI, etc.

Contact

- Email: xianggu@xjtu.edu.cn
- Office: No. 434, Building of Mathematics, Xingqing Campus, Xi'an Jiaotong University.
- Address: No. 28, Xianning West Road, Xi'an, Shaanxi 710049, P. R. China.